Mapping sequencing reads

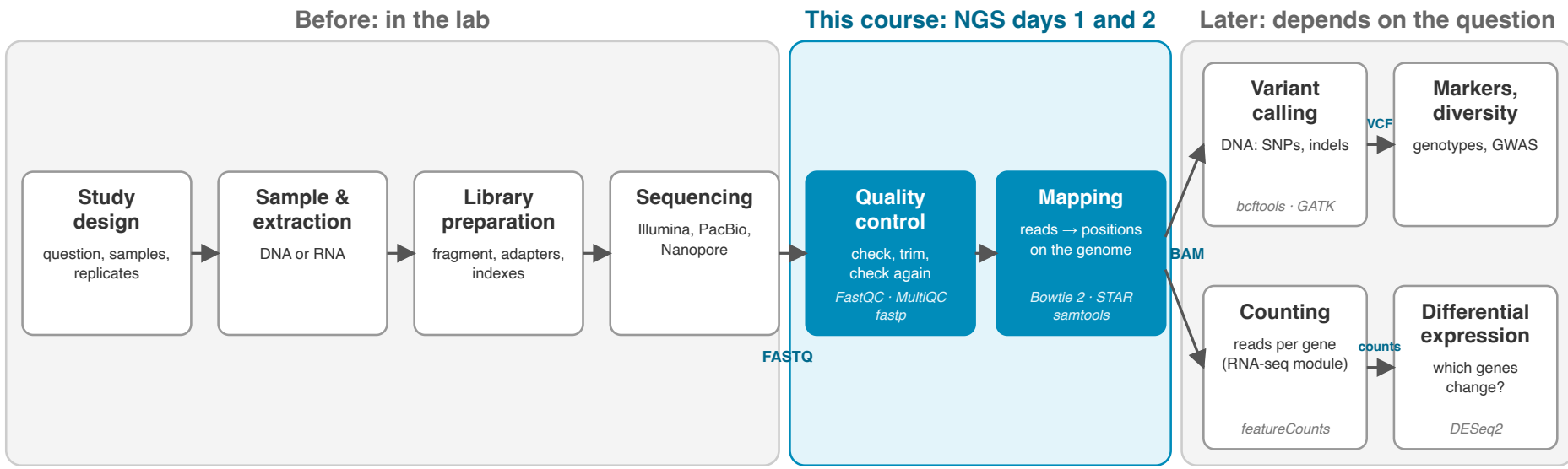
Introduction to Bioinformatics · NGS day 2

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This lecture

1. Mapping and BLAST: what they do, and how they differ
2. Alignments: mismatches, gaps, pairs and repeats
3. RNA: reads across splice junctions
4. SAM files and mapping statistics
5. The exercises

Where we are in the workflow



What is mapping?

From reads to positions

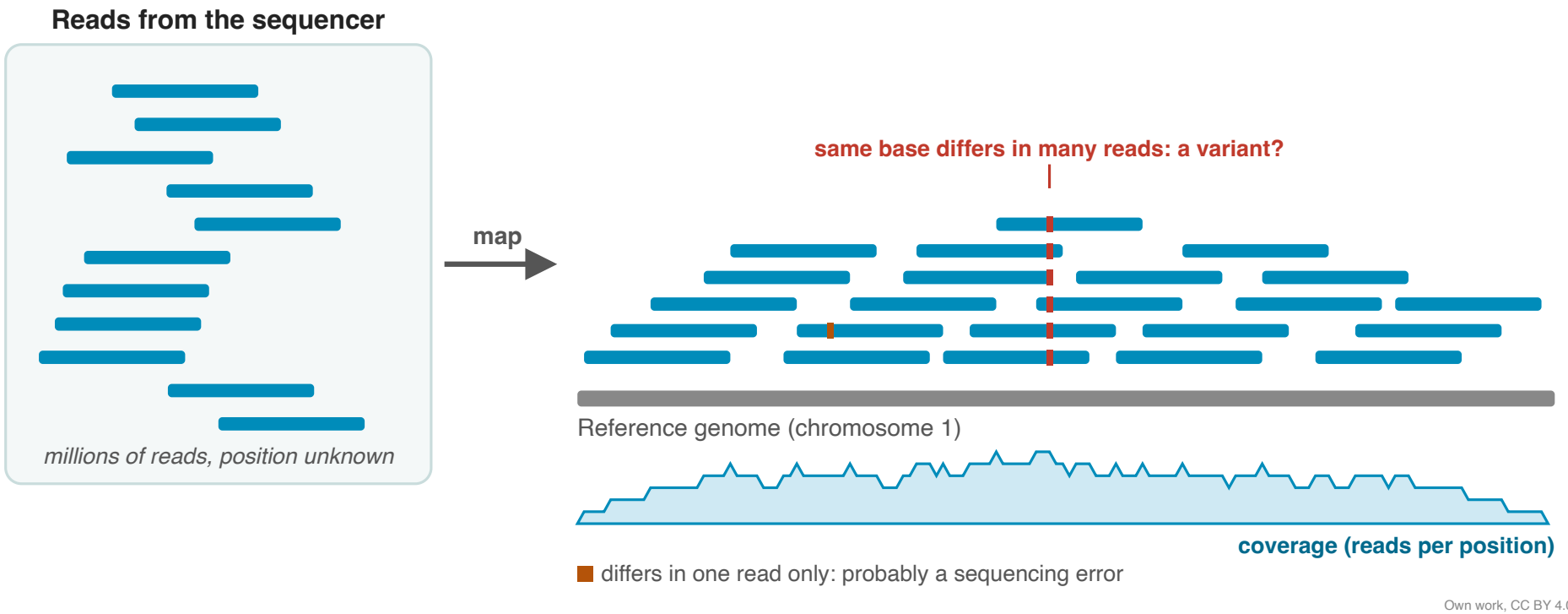
After sequencing we have millions of short reads, and we don't know where any of them came from.

Mapping finds, for each read, the place in a **reference** where it fits best.

- **Reference genome:** the assembled chromosomes of the species
- **Reference transcriptome:** the sequences of all known transcripts

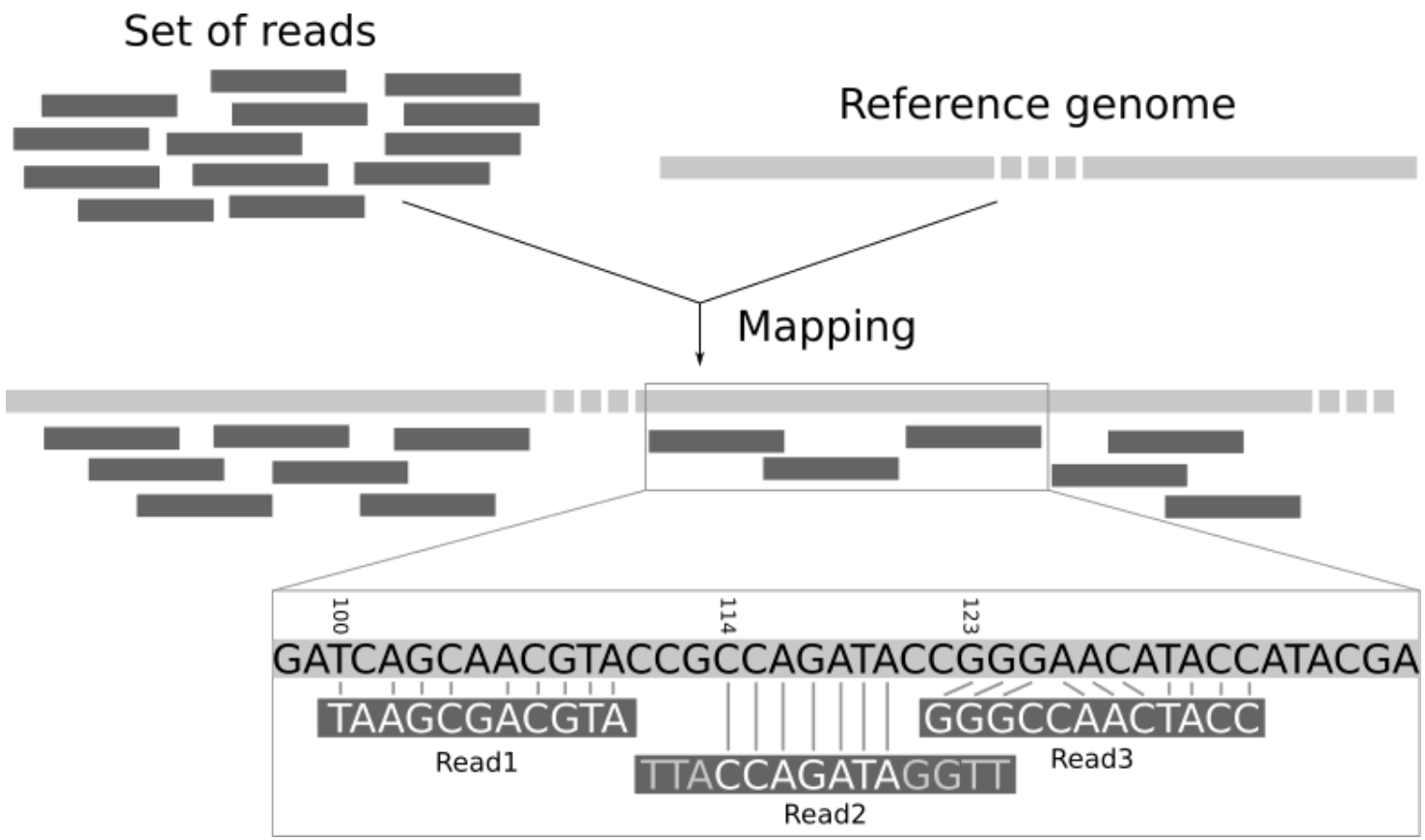
Once reads have a position we can count them per gene, look for variants, or check coverage.

Many reads, one reference



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Mapping in a picture



Read 1: two mismatches. Read 2: local alignment, both ends clipped. Read 3: a 2-base insertion and a 1-base deletion. Figure: Galaxy Training Network

BLAST: searching for similar sequences

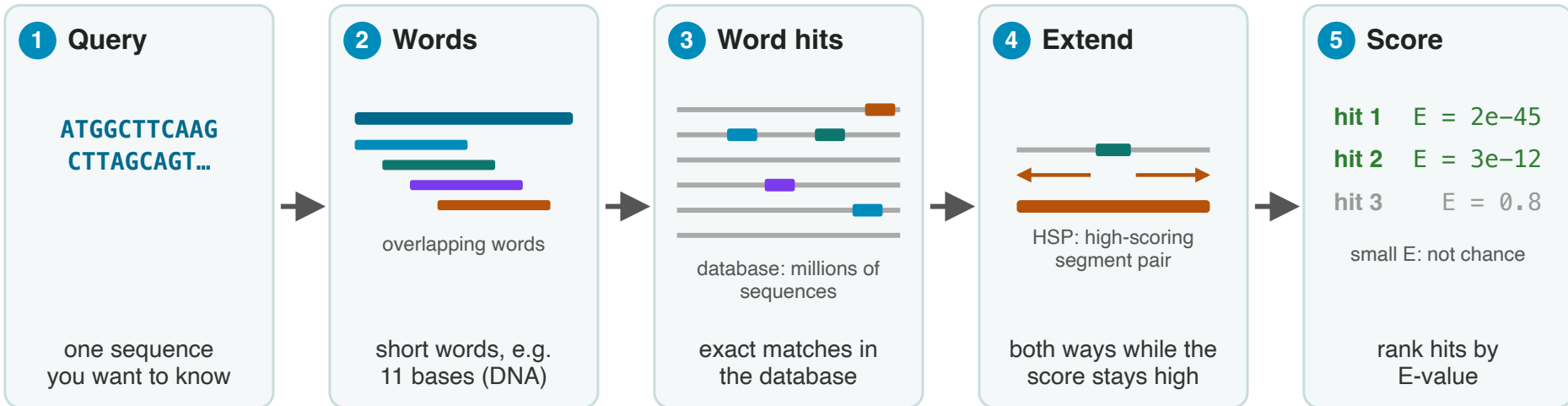
BLAST

Basic **L**ocal **A**lignment **S**earch **T**ool (Altschul et al., 1990)

- One sequence, the **query**, against a large **database** (for example NCBI nt, or all proteins)
- Finds **similar** sequences, not only identical ones, also in other species
- Answers: *what is this sequence?*

Program	Query	Database
blastn	DNA	DNA
blastp	protein	protein
blastx	DNA, translated	protein
tblastn	protein	DNA, translated

How BLAST works



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Seed and extend: look for short exact matches first, then align only around them.

Reading a BLAST hit

Column	Meaning
% identity	Share of identical positions in the alignment
Alignment length	How much of the query is aligned
Bit score	Alignment score, independent of the database size
E-value	Number of hits this good expected **by chance** in a database this size

E-value: close to 0 (e.g. 1e-50) means the hit is very unlikely to be chance. Around 1 or above means it could easily be chance.

BLAST or a mapper?

	BLAST	Read mapper (Bowtie 2, STAR)
Question	What is this sequence?	Where in **this** genome does it come from?
Queries	One or a few	Millions of reads
Database	Many species	One reference genome
Differences allowed	Many: finds distant relatives	Few: same or closely related species
Speed per sequence	Seconds	Microseconds

Both **seed and extend**. Mappers make it fast by indexing one reference **once**, and by expecting the read to be almost identical to it.

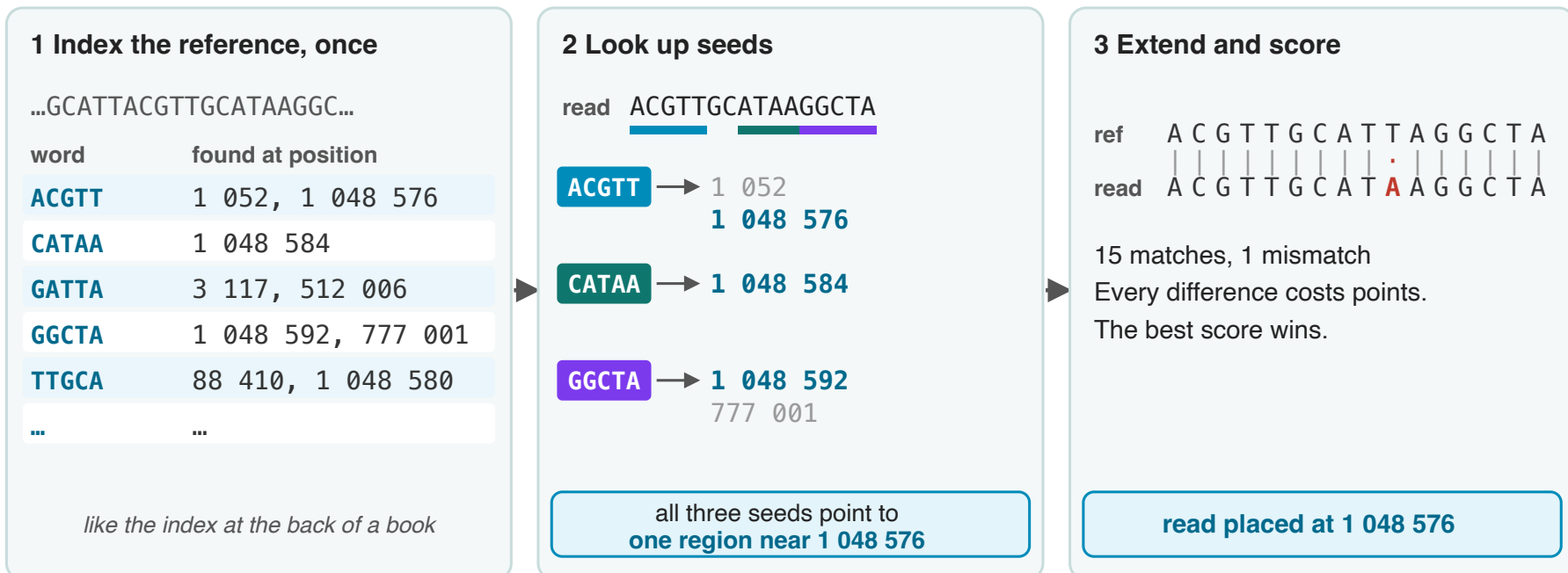
Finding the place fast

Why not just search?

- 20 million read pairs × a genome of roughly 700 Mb (cassava)
- Comparing every read with every position: about 10^{16} comparisons
- Searching with BLAST would take weeks

Mappers first build an **index** of the reference, once. Then each read is looked up in the index, like a word in the index at the back of a book.

Index, seed, extend



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Each mapper builds its **own** index format. A Bowtie 2 index cannot be used by STAR.

Alignments

What an alignment allows

Difference	Why it happens
Mismatch	Sequencing error, or a real variant (SNP)
Insertion / deletion	Sequencing error (homopolymers), or a real indel
Soft clipping	Adapter left on the read, low-quality ends, structural variant

Every difference costs points. The mapper reports the alignment with the best score.

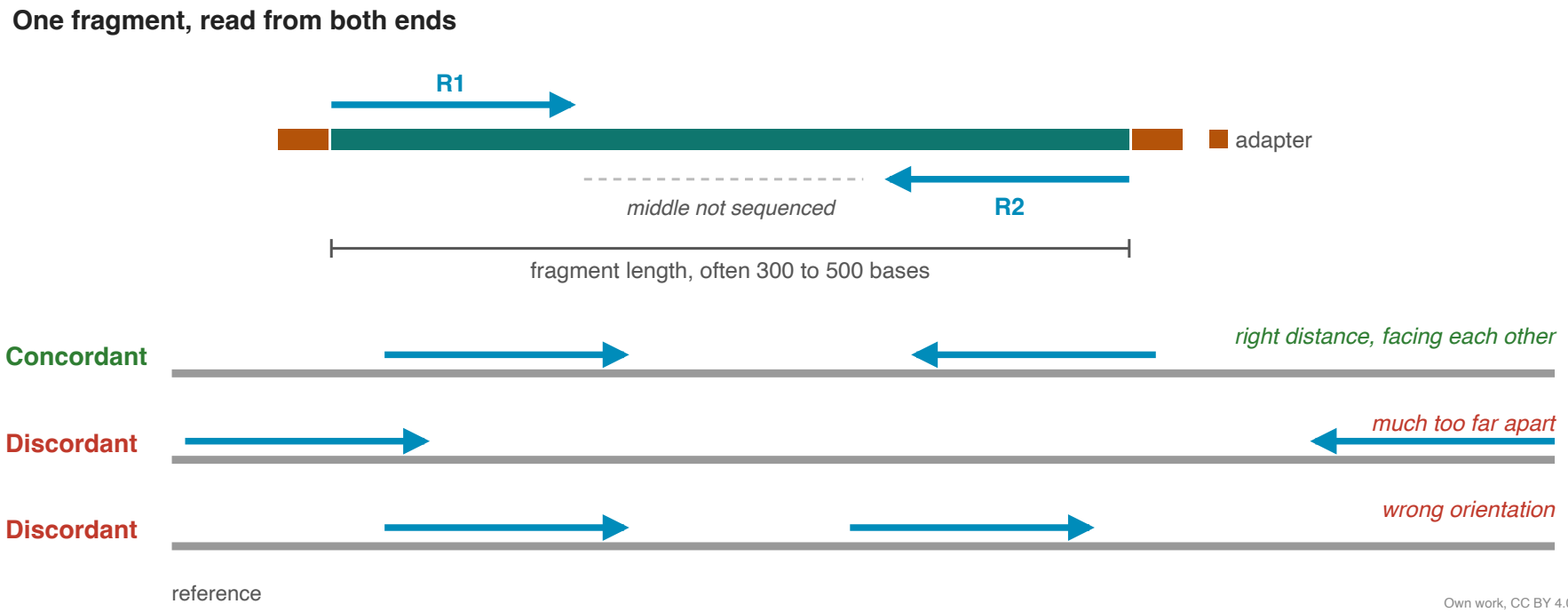
End-to-end or local?

End-to-end: the whole read must align. Unaligned ends are penalised as mismatches.

Local: the ends may be **clipped** if they don't fit. Good when reads still carry adapter or low-quality ends.

In Bowtie 2: `--very-sensitive` (end-to-end) vs `--very-sensitive-local` (local). You will try both.

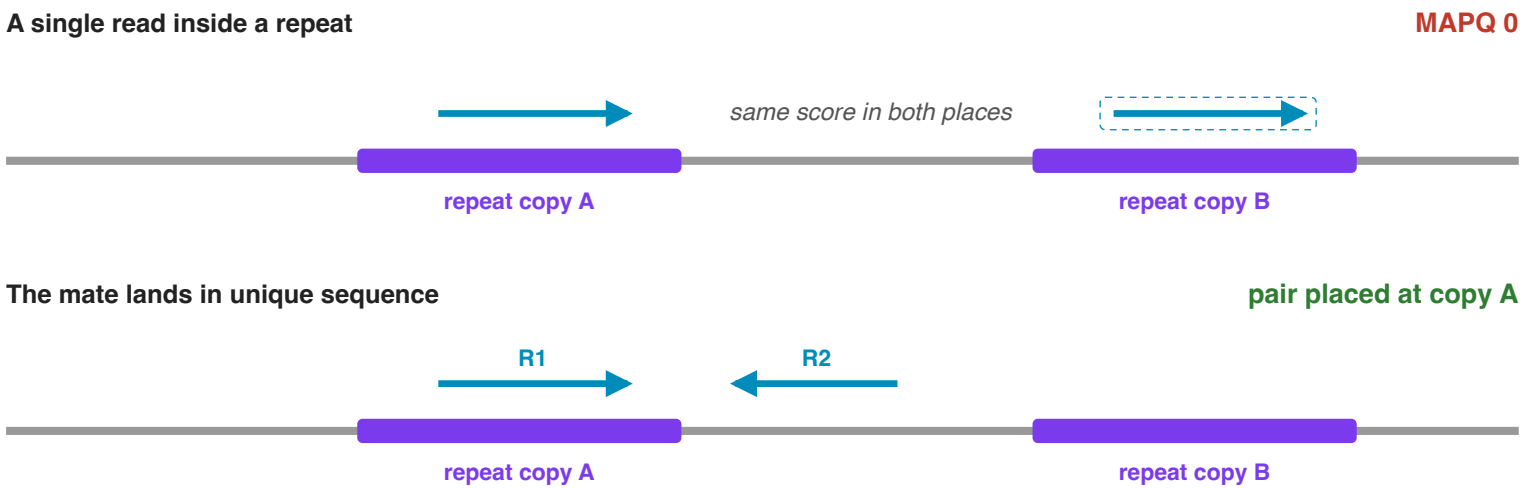
Paired-end reads



A pair that maps as expected is **concordant** (Bowtie 2) or **properly paired** (samtools).

Reads that fit in many places

Genomes are full of repeats: transposons, gene families, duplicated regions.



- A read that fits equally well in several places is a **multi-mapping** read. The mapper picks one at random, or reports all of them.
- **MAPQ** (mapping quality) says how sure the mapper is: 0 means it could be anywhere.

Plant genomes are special

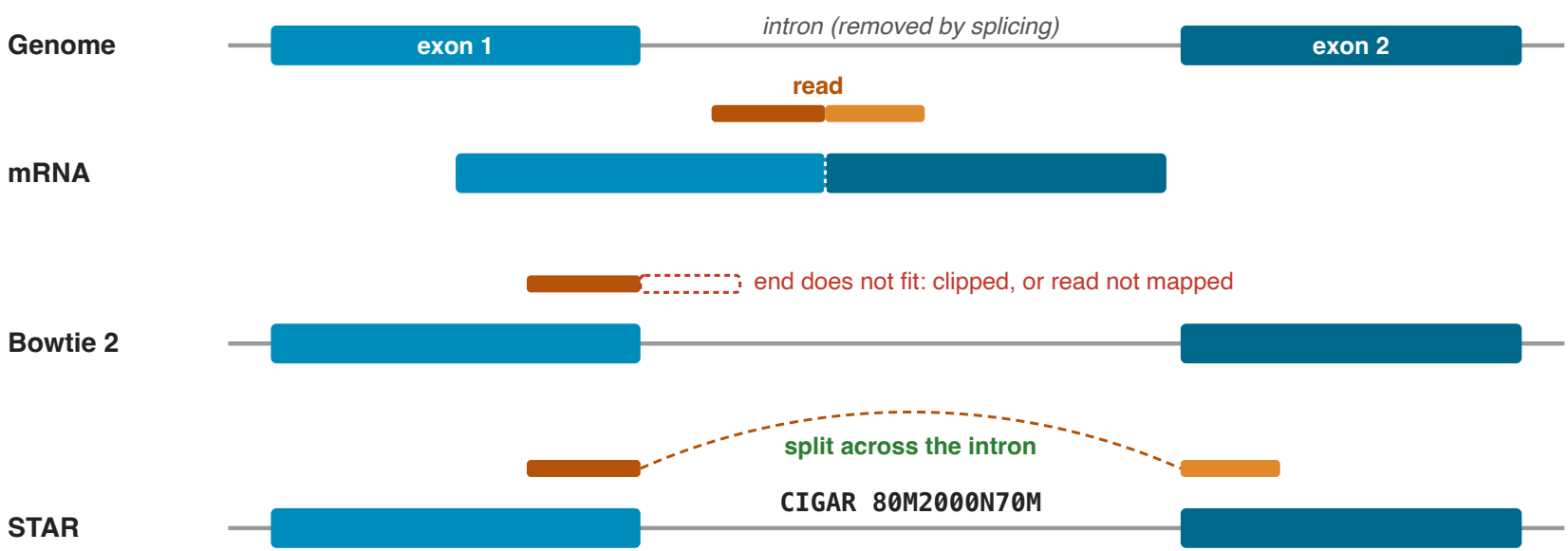
- Large and full of **transposable elements**: often more than half the genome
- Often **polyploid**: several copies of each chromosome, similar but not identical
- Often **heterozygous**: cassava is, so the two copies of a chromosome differ

Your cultivar is not the reference cultivar. Differences between them will look like mismatches to the mapper.

RNA

Reads across exons

In RNA-seq, reads come from **spliced** transcripts. Many reads span an exon-exon junction.



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Splice-aware mappers

- **STAR**, **HISAT2**: allow a read to be split across an intron
- An annotation (**GTF**) tells them where known exons and junctions are
- They also find **new** junctions from the data

That is why STAR needs the `.gtf` file when it builds its index, and Bowtie 2 doesn't.

Which mapper for which data?

Data	Typical tools
DNA, short reads	Bowtie 2, BWA-MEM2
RNA, short reads	STAR, HISAT2
Long reads (DNA or RNA)	minimap2
RNA, counts only	Salmon, kallisto (pseudo-alignment, no exact positions)

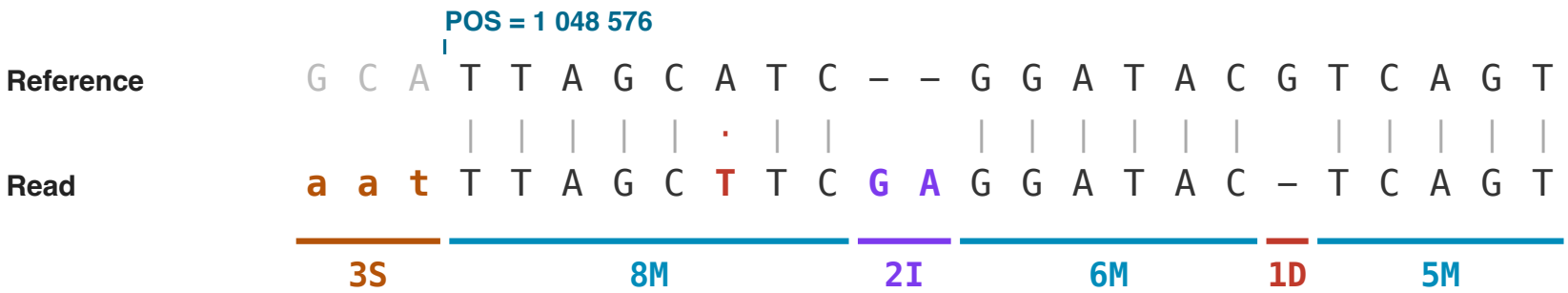
The SAM format

SAM: one line per alignment

A header (lines starting with `@`), then 11 tab-separated columns per read:

Column	Name	Example
1	Read name	`read_123`
2	FLAG	`99`
3	Reference	`Chromosome01`
4	Position	`1048576`
5	MAPQ	`42`
6	CIGAR	`5S70M`
7 to 9	Mate: reference, position, fragment length	`=`, `1048790`, `289`
10, 11	Sequence, qualities	`ACGT…`, `FFF:…`

CIGAR: read alignment



CIGAR `3S8M2I6M1D5M`

S soft-clipped, not aligned
M aligned (match or mismatch)
I inserted in the read · D deleted from the read

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In RNA-seq you will also see `N`: a skipped region, the intron (e.g. `80M2000N70M`).

FLAG: yes/no answers in one number

FLAG is a sum. Each yes/no answer has its own value:

Value	Meaning	In FLAG 99?
1	Read is paired	yes
2	Pair is properly paired	yes
4	Read is unmapped	
16	Read is on the reverse strand	
32	Mate is on the reverse strand	yes
64	First read of the pair (R1)	yes
128	Second read of the pair (R2)	
256	Not the primary alignment	

99 = 1 + 2 + 32 + 64. Decode any FLAG at broadinstitute.github.io/picard/explain-flags.html.

SAM, BAM and samtools

- **SAM** is text: readable, but large
- **BAM** is the compressed binary version of the same data
- **samtools** converts, sorts, indexes and summarises them

```
samtools view -b sample.sam > sample.bam
samtools sort -o sample.sorted.bam sample.bam
samtools index sample.sorted.bam
samtools flagstat sample.sorted.bam
```

Mapping statistics

What does a good mapping look like?

	DNA, same species	RNA-seq
Overall alignment rate	above 90 %	70 to 90 % uniquely mapped
Properly paired	most pairs	most pairs
Multi-mapping	a few %	a few % to 10 %

These are rules of thumb for a **whole** reference genome.

Unmapped reads

- Wrong or incomplete reference
- Contamination (bacteria, fungi, human)
- Adapters or poor quality left on the reads
- A divergent cultivar or species
- rRNA in RNA-seq

Important

In the exercise we map to **chromosome 1 only**. Cassava has 18 chromosomes. What mapping rate do you expect?

The exercises

You will

Sample	Data	Mapper	Index needs
TMEB117	DNA, paired-end	Bowtie 2	chromosome01.fasta
TMEB419	RNA, paired-end	STAR	chromosome01.fasta + .gtf

- Build both indexes, map, read the logs
- Compare raw and trimmed reads, local and end-to-end
- Look inside the SAM file with samtools

Summary

- Mapping places each read on a reference, using an index to be fast
- BLAST finds similar sequences across species; mappers place millions of reads in one genome
- Alignments allow mismatches, gaps and clipping, each at a cost
- Pairs and MAPQ help with repeats; plant genomes have many
- RNA needs a splice-aware mapper such as STAR
- SAM/BAM stores the result; the log tells you if it worked